Service lines

Seven lines in three groups, with one team responsible from start to finish.

58	7	4
YEARS · 1968	COUNTRIES WITH PROJECTS	OFFICES

Building industry with people who deliver, since 1968.

GRUPO HL, FORMERLY HL INGENIEROS · HL INGENIEROS S.A. · HLINGENIEROS.COM

EPC and turnkey

A · MANAGEMENT

We deliver the complete project, from engineering to commissioning, handing over a fully operational production plant.

Scope

Basic and detailed engineering

Basic and detailed engineering in the civil, mechanical and electrical disciplines, supporting the execution of our construction projects for every industrial sector.

Procurement

Management of project purchasing and supply as part of the full integration of its phases.

Construction and installation

Civil works and electromechanical installation for the project.

Commissioning and start-up

Commissioning, testing and support to the operating staff until the investment project becomes a fully operational plant.

Resources and partners

Basic and detailed engineering is carried out through partners and contractors; the turnkey model is backed by strategic alliances.

Standards

NSR-10

Reference projects

- **Columbus Plant** · Cementos Argos · Cartagena, Colombia

Project management

A · MANAGEMENT

We plan, schedule and control every project as a whole, from concept to commissioning.

Scope

Project management (PMC and EPCM)

Our teams take the client's requirements and integrate every knowledge area to plan, schedule and control the project, from conceptual design through execution and commissioning.

Cost control

Cost control with early warnings and periodic reports for decision-making.

Technical control

Schedule updates, measurement of actual and contractual physical progress, productivity and S-curve.

Contract administration

Support in managing and administering subcontractors, progress certificates and contract control, with alerts for scope and schedule deviations.

Risk management

Identification and qualitative analysis of risks and success factors, root cause analysis, response actions and owners.

Inspection and oversight

Technical, administrative, HSE, environmental and quality inspection and oversight.

Owner's team support

Our specialized staff joins the client's organization, advising on and taking part in the execution of its projects with experience from major construction works.

Training and safety model

Training in the occupational health and safety management system (workshops, seminars and a certificate program) and design and implementation of industrial safety management models.

Reference projects

- **Cementos Diamante Lines 1 and 2** · Cementos Diamante
- **Chuchupa A and B Platforms** · Chevron

Industrial civil works

B · CONSTRUCTION AND INSTALLATION

We directly execute the civil works for industrial plants and for public and private projects.

Scope

Earthworks

Excavation and backfill.

Foundations and concrete structures

Foundations for all types of equipment and buildings, concrete structures and leveling slabs.

Masonry

Masonry for industrial buildings.

Slipform construction, piling and site development

Slipform construction of buildings and silos, piling and site development works, which round out our full civil offering.

Resources and partners

Piling, slipform construction of buildings and silos, and site development works, through strategic alliances.

Standards

NSR-10

Reference projects

- **Columbus Plant** · Cementos Argos · Cartagena, Colombia
- **Termonorte Thermal Power Plant** · Hyundai Heavy Industries · Santa Marta, Colombia
- **GACE IV Project** · Chevron Petroleum Company · Campo Ballenas, Colombia
- **Chimborazo Cement Plant** · UTE Chimborazo (Sacyr, Imasa, Polysius) · Riobamba, Ecuador

Mechanical and piping installation

B · CONSTRUCTION AND INSTALLATION

We install process equipment and its structures to build new plants or upgrade existing production lines.

Scope

Process equipment

Installation, alignment and calibration of process equipment.

Steel structure

Erection of all supporting steel structures.

Coatings, insulation and refractories

Application of coatings, insulation and refractories.

Piping

Prefabrication and installation of piping in every diameter and in industrial materials (carbon steel, stainless steel, HDPE, among others), with the welding process suited to each client's requirements.

Standards

ASME · API 650 · AWS

Reference projects

- **El Salitre WWTP, Phases 1, 3 and 4** · Consorcio PTAR Salitre · Bogotá, Colombia
- **Columbus Plant** · Cementos Argos · Cartagena, Colombia
- **New Cementos Progreso Plant** · thyssenkrupp Industrial Solutions · San Juan Sacatepéquez, Guatemala
- **Buriticá Process Plant** · Continental Gold · Buriticá, Colombia

Electrical installation and instrumentation

B · CONSTRUCTION AND INSTALLATION

We install the electrical power and control system and the instrumentation needed to commission each project.

Scope

Power

Medium- and low-voltage installations, electrical substations, distribution boards, switchgear and motor control centers (MCC).

Grounding and lightning protection

Grounding grids and lightning protection.

Cabling and raceways

All required cabling and raceways.

Instrumentation and control

Installation of instrumentation for integrated control of the production process.

Standards

RETIE

Reference projects

- **Columbus Plant** · Cementos Argos · Cartagena, Colombia
- **Buriticá Process Plant** · Continental Gold · Buriticá, Colombia
- **Cementos Diamante Lines 1 and 2** · Cementos Diamante
- **GACE IV Project** · Chevron Petroleum Company · Campo Ballenas, Colombia

Metal fabrication

C · FABRICATION AND MAINTENANCE

We fabricate capital goods for industry, choosing the best solution for each specialty, the logistics and the trade agreements of each country.

Scope

Steel structures

Steel structures for industry.

Tanks and vessels

Atmospheric and pressure tanks.

Boilers and ducts

Boilers and ducts.

Material handling

Material handling equipment, among other capital goods.

Resources and partners

We develop partners in every region where we operate, according to the fabrication specialty, transport logistics and the trade agreements between the countries in the area.

Standards

ASME · API 650 · AWS

Reference projects

- **El Salitre WWTP, Phases 1, 3 and 4** · Consorcio PTAR Salitre · Bogotá, Colombia
- **Cementos Diamante Lines 1 and 2** · Cementos Diamante
- **Ladrillera Santa Fe Block Plant** · Ladrillera Santa Fe

Maintenance and plant shutdowns

C · FABRICATION AND MAINTENANCE

We carry out major maintenance of production plants and treat every shutdown as a project with very tight deadlines.

Scope

Plant shutdowns

Major maintenance of production plants, conceived as highly demanding projects with very short time limits.

Routine maintenance

Field staff who reinforce the client's maintenance team in routine work.

Reference projects

- **Cerrejón Shiploader Repair** · Carbones del Cerrejón
- **Caracolito Baghouse** · Cemex